

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm
Duration: 1 hour 30 min

Semester: Summer 2025
Full Marks: 50

CSE 320: Data Communications

Answer the following questions [Total 3].
Figures in the right margin indicate marks.

SET B

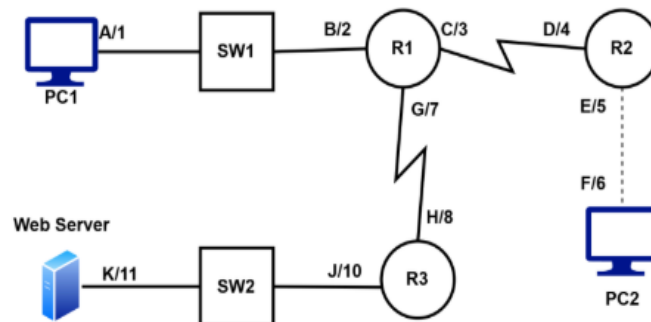
Name:	ID:	Section:
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1. a) Suppose a dormitory has five buildings. Each of the buildings has three sections as follows: library, [3+2]
[CO1] work station and break room. Each section has nodes as follows: four nodes for the library, 6 nodes for the work station and 2 nodes for the break room. Now, **draw** the hybrid topologies according to the following configuration, **labelling** the sections:

- The nodes in each section have a dedicated connection to all other nodes.
- The buildings are connected to the neighboring buildings only.
- The sections are connected directly to a device which is a central point of failure. Meaning if it goes offline, then all the connected devices go offline.

Based on the above topology, calculate the total number of connection links.

- b) PC2 using port number 49543 has sent a request packet to a Web server (port number 443). The [3]
packet is currently at router R1, moving towards the server. **Identify** the source and destination Port, IP, and MAC Addresses of this request packet. Consider the alphabets as physical addresses and the numbers as logical addresses (given beside the devices in the figure).



- c) Imagine you are a network engineer working for a company that needs to send a message from [4+2]
Computer A to Computer B over a network. The message is: **"Hello, TCP/IP!"**

Your task is to break down how this message is processed at each layer of the TCP/IP protocol suite. You need to:

- For each layer, **identify** the specific name of the data unit and the type of addressing used at that layer.
- Illustrate** how headers and trailers are added as the data moves through the layers.

2. [CO2] a) A video streaming company is designing a network to stream 1080p HD video (1920x1080 pixels, 120 fps, 24-bit color) to users in a rural area. The network uses a wireless communication channel, which introduces some noise due to interference and long-range transmission. To ensure faster transmission, the video's raw bitrate will be reduced by a factor of 80 before final delivery. [4+3+3]

Wireless Channel Specs:

- Channel Bandwidth = 10 MHz
- Signal-to-Noise ratio = 30 dB

Hint: FPS means frames per second; each frame has the given resolution, and each pixel in that resolution uses the given color depth in bits.

You as an engineer need to determine:

- What is the **bitrate** required to stream the video considering the compressing?
 - What is the number of **signal levels** required for the stream assuming a noiseless channel. **How** many bits do you need to send per signal.
 - What is the maximum **achievable bitrate** over the actual noisy channel? Can the video stream be transmitted reliably over the current channel?
- b) If the signal-to-noise ratio (SNR) is close to infinity for a communication channel, what can be said about this channel? **Mention** some causes of transmission impairment. [2+1]
- c) A non-periodic composite signal contains frequencies from 5 to 40 KHz. The peak amplitude is 10 V for the lowest and the highest signals and is 30 V for the 20-KHz signal. If the amplitudes change gradually from the minimum to the maximum and then maximum to the minimum, **draw** the frequency spectrum. [3]
- d) **Distinguish** between data transmission delay and propagation delay. [1+3]

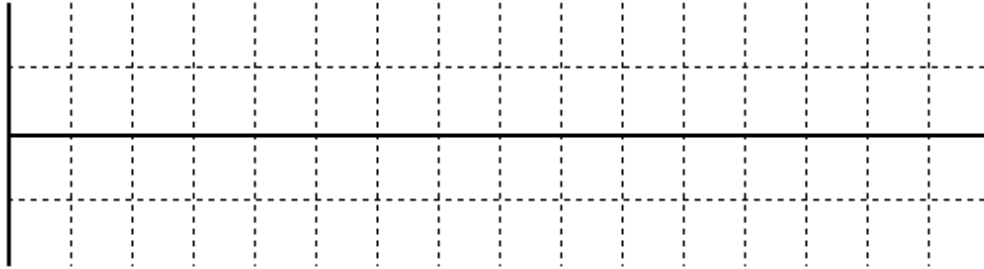
If a signal at the beginning of a cable with **-0.65 dB/km** (the loss in a cable is usually defined in decibels per kilometer) has a power of 10.5 mW, **calculate** the power of the signal at 5.2 km.

3. [CO2] a) Suppose you are applying 3B/5B block coding in your network system. To get certain advantages, you are following some rules to map the 3-bit combinations with the 5-bit combinations. Observe the below mapping table - [3+1]
- **Write** down the rule/s that are applied to make a 5-bit codeword valid in the given table.
 - If you are transferring the data at a speed of 200 kbps as a sender, then **what** will be the **transmission speed** after applying the 3B/5B substitution?

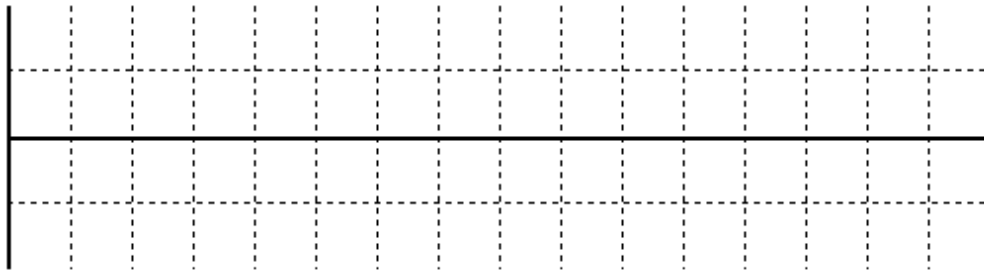
3-bit input	5-bit codeword
000	11010
001	11110
010	10110
011	10011
100	01011
101	01110
110	10101
111	11001

- b) The NRZ-I scheme does not solve the problem of DC components. But, if the DC component problem is unacceptable, then **what scheme** can be used alternatively, which holds $r = 1/2$ (ratio of data element and signal element)? [1+2]

For the data stream 011100001000001, **apply** the line coding scheme that you are choosing.



- c) Now solve the long sequence of 0's problem in the above data stream given in 3(b) using a suitable method, keeping $r = 1$ (ratio of data element and signal element). [3]



- d) The following table depicts a sampled low pass analog signal with a bandwidth of 200 KHz. [4+2]

Time	Amplitude (v)
0	11.5
1	-28.3
2	62.5
3	-45.8

By applying the concept of **Pulse Code Modulation** for analog to digital conversion, **show** the normalized quantized values and encoded words for the given amplitudes at different time stamps. The sampling amplitudes are between -27V to +69V and there will be 8 levels in total.

